

It's All Fun and Games, Until Someone Gets Hurt: How Game Pace and Player Movement Influence the User Experience in VR

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Figure 1: Screenshots from our customized VR game showing the two types of weapons used to elicit differences in game pace and user movements while keeping the game modes comparable.

Abstract

Affordable stand-alone HMDs make VR more easily accessible, but a surge in VR-related injuries highlights persistent safety risks. Awareness of these risks may lead to fear of colliding with real-world obstacles and distract from the VR application. To examine the link between game design and collision anxiety, we created two versions of a VR game: a slow-paced variant with minimal movement and a fast-paced version requiring extensive motion. The slower game elicited significantly less collision anxiety in a within-subject design ($N = 35$). However, sequencing effects emerged: participants who first played the fast-paced game reported higher anxiety during the slower game than those who started with the slower version. These findings suggest that while movement intensity influences collision anxiety, it also depends on user expertise and priming from prior experiences. Our study underscores the need for further research to understand the impact of collision anxiety on VR user experience.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; Virtual reality; • **General and reference** → Empirical studies.

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Keywords

virtual reality, user experience, user expertise, collision anxiety, discomfort, assessment, questionnaire, spatial orientation, VR orientation, VR games

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1 Introduction

Virtual Reality (VR) has gained significant attention in recent years, both in industry and academic research. Driven by recent technological advancements from companies like Meta, HTC, and Pico, VR headsets have become more affordable and accessible for home use. As a result, sales of head-mounted displays (HMDs) have surged in recent years¹. Stand-alone HMDs without a cable allow for a greater range of movement and higher levels of immersion. However, this also introduces challenges: The greater freedom of movement carries the risk of real-world collisions, especially in smaller, cluttered private homes [5, 8, 19].

One popular use case for private VR use are exergames in VR, i.e., games that require body movement to control the game [14]. VR exergames might pose a solution to an increasingly sedentary lifestyle of today's world, as the high immersion and spatial nature

¹<https://arinsider.co/2022/10/26/will-vr-revenue-exceed-28-billion-by-2026>, last visit: 22.01.2025

of VR games can help increase motivation [10]. However, especially with VR exergames that inherently require a lot of large, fast movements, collisions with room boundaries, objects, or bystanders become more probable. To avoid collisions, the player might consciously or subconsciously focus on avoiding collisions instead of fully engaging with the game. This shift in attention might diminish the overall player experience [16].

The fear of real-world collisions during the use of VR applications has been defined as *collision anxiety* and is a relatively new field of research [16]. As such, there is a lack of fundamental research regarding the origin and emergence of collision anxiety. However, a better understanding of these factors can guide the design of future VR applications and safety systems. Two variables that may fundamentally impact collision anxiety are the pace of the game and the extent of movement required from the player. Larger movements carry a higher risk of hitting objects or losing orientation while fast movements make potential collisions more dangerous. While it seems plausible that large movements and fast-paced games are associated with a higher risk of collisions and, hence, higher collision anxiety, no research has been done to prove or disprove this hypothesis. Hence, our work focuses on the research question:

RQ: Does a faster game pace and greater extent of movement required from the player in a VR game lead to greater collision anxiety?

To answer our research question, we have developed two versions of a game that differ in game pace and required player movement. By contrasting collision anxiety and overall player experience in both games, we gained insights into the relationship between player movement and collision anxiety. Additionally, we uncovered sequencing effects that hint towards further influences of collision anxiety on the player experience. While research remains to be done in this field, we hope to offer important basal insights to enable a better understanding of the relationship between application design and collision anxiety.

2 User Experience in VR

VR offers a highly immersive experience, which can be beneficial for the user experience. However, especially in unsuitable spaces, users risk a collision with real-world obstacles or bystanders [5, 8]. While all commercial HMDs come with safety or chaperone systems, such as Meta's Guardian, the high number of VR accidents suggests that safety systems do not eliminate the risk of accidents [4, 5]. Cucher et al. [4] examined VR-related injuries between 2013 to 2021. While the first VR-related injury was reported in 2017—along with 124 more in that year—there were already 1,336 VR-related injuries in 2021.

2.1 Collision Anxiety

Awareness of this risk might manifest as collision anxiety, the *feeling of discomfort, anxiety, or fear of colliding with or hitting the real-world environment or another person while using an XR application* (p.3) [16]. While the subjective feeling of safety or fear of real-world collisions has been mentioned in earlier studies, it has only been systematically assessed using the collision anxiety questionnaire (CAQ [2, 12, 13, 16]). Boletsis & Cedergren [2] examined the walking-in-place locomotion technique and reported that fear

of real-world collisions caused a break in the participants' immersion. Similar, work by Mousas and colleagues [13] hints towards a physiological indication for collision anxiety: They found that when the real environment did not match the virtual one, participants showed physiological arousal and a changed movement behavior along with reported worries about colliding with real-world obstacles. Supplementary to this, a study by Liu and colleagues [12] designed an approach for virtual environments to represent real-world obstacles. Their work showed that virtual environments that represent real obstacles as virtual objects improved the participants' presence and reduced the fear of colliding. One reason for accidents related to VR is suggested by Kanamori et al. [9], who showed that with the current chaperon systems, users might not understand their distances to real-world objects. This is even more important in multi-player games, where users might feel anxiety when at risk of colliding with another person when sharing a play space. In a study by Podkosova & Kaufmann [15], users reported a focus shift away from the application and towards avoiding collisions when sharing their play space with another user compared to having their own space. Overall, research suggests that collision anxiety can distract users from the application and might decrease the user experience. This concludes that understanding collision anxiety is an essential step toward improving the player experience.

3 Research Method

To address the research question of how game pace and extent of movement influence collision anxiety, we designed a VR game in two variations: one with a slower pace and less required movement and the other with a faster pace and more extensive movement. We used a within-subject design, meaning that each participant played both game versions.

3.1 Game Development and Versions

We chose to develop our own VR game—played on a Meta Quest 2—with the game engine Unity² to be able to tailor the game precisely to our research objective. The game's theme revolves around a ghost-hunting scenario set in a haunted castle, where players take on the role of a ghost hunter tasked with eliminating the ghosts, sequentially with two pistols or two swords. The virtual environment mirrored the measurements of the physical room where the study was conducted (3x4 meters), aligning virtual walls with real-world boundaries. The game took around five minutes to complete, and players were guided through a tutorial before starting the game.

3.1.1 Differences between Game Versions. One of the main distinctions between the two game versions is the type of weapon used. In the faster-paced, higher-movement version, players wield swords to combat the ghosts. As short-ranged melee weapons, swords require the players to move through the room to engage the enemy and make large upper-body movements to fight the ghosts. In contrast, the slower-paced, less movement-extensive version provides players with pistols, allowing them to play more stationary, as they only have to take aim and press the trigger button to shoot. After the pistol has released a bullet, it needs three seconds to recharge, intentionally slowing down the gameplay. Once the pistol

²<https://unity.com>

has recharged, it is glowing, as seen in Figure 1. We also adjusted the enemy behavior: In the sword version, ghosts spawn in a 360° area surrounding the player and at various heights, which requires players to turn and reach further to fight the ghosts. By contrast, in the pistol version, ghosts spawn only within a narrow area in front of the player and at eye level, allowing players to engage while stationary and without needing to turn or walk around the room (see Figure 1). Additionally, the two versions differ in terms of the number of ghosts that spawn simultaneously and overall, with three simultaneous ghosts spawning in the sword version and only two in the pistol version. Last, the ghosts move faster in the sword version, forcing the players to make faster movements to successfully hit the ghosts.

3.2 Measurements

We measured collision anxiety, perceived exertion, player experience, presence, and cyber sickness to identify potential differences between both versions.

3.2.1 Exertion and Cyber Sickness. To measure perceived exertion, the Borg Scale was used [3], which rates exertion on a scale from 6 to 20 and estimates the actual heart rate as roughly ten times the selected Borg value. Additionally, the Fast Motion Sickness Scale (FMS) was applied to assess cyber sickness [11]. The FMS is a single-item scale ranging from 0 (no sickness at all) to 20 (severe sickness).

3.2.2 Collision Anxiety. Collision anxiety was evaluated using the validated German version of the Collision Anxiety Questionnaire (CAQ, [16, 17]). The CAQ consists of 10 items distributed across three subscales: general collision anxiety (GCA), orientation (ORI), and interpersonal collision anxiety (IPC). Since no interpersonal collisions were possible, we followed the authors' advice and only used the GCA and ORI subscale.

3.2.3 Player Experience and Presence. General player experience was evaluated using the Mini Player Experience Inventory (miniPXI, [7]). This shortened version of the original Player Experience Inventory by Abee et al. [1] has only 11 items measuring the player's gaming experience on a 7-point Likert scale, ranging from 'strongly disagree' to 'strongly agree'. We used the German translation for the PXI items by Graf et al. [6]. The player's sense of presence was measured using the IGroup Presence Questionnaire (IPQ, [18]). The questionnaire includes four subscales: general presence, spatial presence, involvement, and experienced realism. Each subscale is rated on a 7-point Likert scale ranging from 0 to 6.

3.3 Pre-Study

To ensure the game versions differed only in movement and game pace but not in other variables, a pre-study was conducted, focusing on perceived exertion, player experience, and the players' collision anxiety. Using a within-subjects design, five participants were instructed to play both game conditions in a randomized order. After each session, they completed the questionnaires listed above. We checked for descriptive trends and calculated t-tests to compare both game versions. However, due to the very small sample size, significant effects would only be expected for a large effect size.

3.3.1 Pre-Study Results. As intended, the perceived exertion differed between both games, with the pistol variant ($M = 9.60, SD = 2.41$) being rated as causing less exertion than the sword version ($M = 10.8, SD = 0.45$). However, the difference was not significant ($t(4) = 1.24, p = .284, d = 0.55$). The same is true for collision anxiety: Here, players experienced more collision anxiety in the sword version ($M = 2.12, SD = 0.80$) than in the pistol version ($M = 1.60, SD = 0.46$), which was again not significant ($t(4) = 1.85, p = .139, d = 0.83$). We did find a significant difference in player experience ($t(4) = 3.76, p = .020, d = 1.68$), which prompted us to make some adjustments to minimize unintended confounding factors. Qualitative feedback indicated that the pistol's bullet speed, was too slow, leading to frequent missed shots. Last, participants expressed confusion about when the pistols had recharged.

3.3.2 Adjustments Based on the Pre-Study. In response to these concerns raised by nearly all five participants as well as the difference in user experience, the bullets' speed was increased while still maintaining the slower pace of the game by keeping the charging mechanic, which was also adjusted. A glow effect was added to indicate when the pistol was ready to fire, as shown in Figure 1. These changes were implemented to reduce the disparity in player experience between the two game versions, enhancing comparability.

3.4 Study Procedure

The study design was approved by the department's ethics committee before data was collected. After a short briefing, we asked participants to fill out demographic questionnaires regarding their gender, age, occupation, and experience with VR. We then randomly assigned all participants to play either the pistol or the sword version. After the first play-through of either version, participants filled out the Borg scale, the FMS, CAQ, iPQ, and miniPXI. Participants were then asked to play the other version and afterward, again answer the Borg scale, the FMS, CAQ, iPQ, and miniPXI. Both game versions, which were capped at five minutes, came with a short tutorial of about one minute. Overall, the study took around 30 minutes, which were rewarded with course credit.

4 Results

Our sample consisted of 35 participants, 21 of which identified as female and 14 as male. The participants' ages ranged from 18 to 28 years ($M = 23.17, SD = 2.64$), and all but three participants were university students. Regarding their VR experience, only eight participants had no previous VR experience at all, but only one participant indicated using VR several times a month. No implausible outliers were detected, and all participants answered the included attention check correctly, allowing the inclusion of all 35 data sets in the analyses.

4.1 Collision Anxiety

The sword version scored significantly higher than the pistol version in all CAQ subscales. Significant differences were visible for the overall CAQ score ($t(34) = -7.95, p < .001, d = -1.34$), the subscale GCA ($t(34) = -9.45, p < .001, d = -1.60$), and the subscale

ORI ($t(34) = -4.12, p < .001, d = -0.70$). Please see the Appendix 1 for all descriptive values per subscale.

4.1.1 Group Differences. An independent sample t-test revealed significant differences in the CAQ score of the pistol version depending on which game version was played first ($t(33) = -3.06, p = .004, d = -1.03$). The group that played the pistol version first had a significantly lower score in this version ($M = 1.78, SD = 0.49$) than the participants who had played the sword version first ($M = 2.42, SD = 0.74$). No such differences were visible in the CAQ score of the sword version between the pistol-first group ($M = 3.22, SD = 0.66$) and the sword-first group ($M = 3.22, SD = 0.58$).

Specifically, when looking at the subscales, participants had significantly higher values in the orientation (ORI) subscale in both game variants when playing the sword version first than when playing the pistol version first (ORI pistol: $t(33) = -3.61, p = .001, d = -1.22$; ORI sword: $t(33) = -2.73, p = .01, d = -.92$; see Appendix 2 for descriptive values).

4.2 Presence

While most iPQ subscales show slight differences between the game versions (see Appendix 3), only the subscale spatial presence shows a significant difference ($t(34) = -2.39, p = .022, d = -0.40$). We calculated Pearson correlations between the subscales of the iPQ and the CAQ. While we did not find significant correlations between the overall CAQ score and the iPQ subscale 'general presence', we found significant positive correlations between the orientation subscale of the CAQ and the involvement subscale of the iPQ for both versions (pistol: $r = .378, p = .025$, sword: $r = .046, p = .005$).

4.3 Borg Scale and Cyber Sickness

There was a significant difference between both game versions in the perceived exertion of the player ($t(34) = -2.78, p = .009, d = -0.47$) with higher values in the sword version ($M = 9.66, SD = 1.61$) than in the pistol version ($M = 8.37, SD = 2.28$). While there was also a significant difference ($t(34) = -3.20, p = .003, d = -0.54$) in cyber sickness between the pistol version ($M = 1.14, SD = 1.73$) and the sword version ($M = 2.29, SD = 2.90$), overall FMS values are still relatively low for both versions.

4.4 Player Experience

The miniPXI shows significant differences between the game versions for the subscales ease of control ($t(34) = -2.77, p = .009, d = -0.47$), mastery ($t(34) = 3.17, p = .003, d = 0.54$), and immersion ($t(34) = 2.80, p = .008, d = 0.47$). Ease of control and mastery were rated higher in the sword version, while immersion was higher in the pistol version. All results can be found in Appendix 4.

5 Discussion

Aiming to understand the relationship between user movement in VR and collision anxiety, we designed two variants of a VR game, casting the player as a ghost hunter in an old mansion. We compared the slow-paced pistol variant, which required little movement, with the fast-paced sword variation, which required larger, fast-paced movements and natural walking to eliminate the ghosts.

5.1 Changes in Collision Anxiety Due to Pace and Movement

The pistol version, which required less player movement, caused significantly less overall collision anxiety than the faster-paced game version. This confirms our research question as the results show that a faster game pace and greater extent of movement required from the player in a VR game might be one cause of increased collision anxiety. While we cannot yet say precisely why there is an observable increase in collision anxiety, the confirmation of a correlation between player movement and game pace and collision anxiety is an important result to build upon.

Participants also reported being significantly more disoriented in the faster version. We assume this to be due to body rotations while fighting the ghosts, as players might lose their orientation in the virtual world and, in turn, in the real world. While the room layout provides a clear conceptual north with two opposite walls featuring either a clearly visible door or window, players might have been too focused on the task at hand. As our sample consisted mainly of inexperienced VR users, they might not have had the mental capacity to keep track of using the VR system, hunting ghosts, and their environment simultaneously. However, future work is necessary to better understand the influence of cognitive load and VR expertise on collision anxiety.

By pre-testing and making adjustments, we aimed to keep both game variants as similar as possible in regard to the user experience so as not to add unwanted confounder variables. However, we did not find a feasible solution to separate large and fast movements without impacting the game's playability, potentially threatening the transferability of our results. Hence, while a combination of both seems plausible, we cannot say whether the speed or extent of movement caused the increase in collision anxiety.

5.2 Sequencing Effects

Besides speed and extent of movement, we also observed sequencing effects to play a role in the emergence of collision anxiety. The sword-first group reported significantly higher collision anxiety in the slower pistol game variant than the pistol-first group. Hence, it seems as if collision anxiety not only stems from the game experience itself but might also depend on user expertise and priming effects from earlier experiences. It might also mean collision anxiety recedes slower than the approximately five to ten minutes it took participants to complete the questionnaires and start the next game variant. This is especially relevant as remaining collision anxiety might act as a confounder variable and spill over into other results in studies with collision anxiety inducing applications.

Participants in the sword-first group were also significantly more disoriented in both game versions than participants of the other group. This suggests that users might subjectively experience collision anxiety as higher if they had no time to familiarize themselves with the surroundings in a high-intensity game. This might hint towards the importance of tutorials or other onboarding measures, especially for inexperienced VR users and high-paced games. While we did include a tutorial level, it focused on the game mechanics and not orientation in the virtual room.

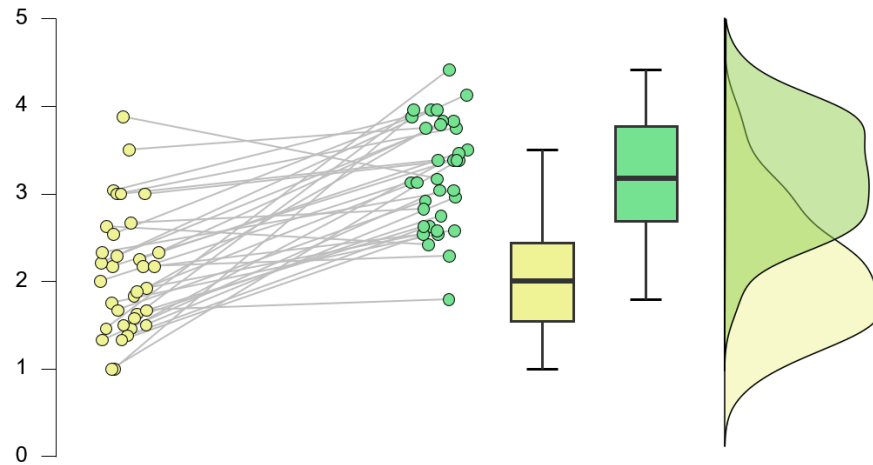


Figure 2: Plots showing the differences in the CAQ score between the slower pistol version (yellow) and the faster sword version (green) of the game. The CAQ values were significantly higher in the sword version of the game.

5.3 Player Experience

While we aimed to keep the player experience as similar as possible between both variations, there were significant differences in three PXI subscales: ease of control, mastery, and immersion. The faster sword version was rated higher in ease of control ('It was easy to know how to perform actions in the game'), implying that users found it easier in this version to interact in the game. This might have been due to the recharge mechanic of the pistols. While we did adjust the recharge mechanic after user feedback in the pre-study, we decided to keep the mechanic as such to ensure a slower pace in this version. The swords seemed to have been more intuitive than the pistols. Participants also rated the sword version higher in mastery, an item asking whether the players felt they were good at playing this game. The higher rating might hint towards a better reception of the more dynamic gameplay of the sword version as opposed to the slower, less engaging pistol version. This, as well as the higher rating in challenge, is especially remarkable as the sword version also elicited significantly higher collision anxiety. It seems as if a more dynamic gameplay might sometimes be worth higher collision anxiety. Finally, immersion ('I was fully focused on the game') was rated higher in the pistol version. However, since the item specifically asks for a focus on the game, we suspect that the higher rating is due to the focus of having to wait for the recharge and carefully taking aim in the pistol version, while the swords can be wielded with less focus. Since we also did not find any correlations between collision anxiety and immersion, we cannot make any statements to the relationship between the two variables.

5.4 Presence

Regarding presence, the main difference between the two versions is the significantly higher spatial presence in the fast-paced sword version. This might be due to differences in movement between the two versions. Participants were encouraged by the game mechanisms to walk through the room and had to turn to engage the

enemies, which might have enabled them to gain a better understanding of the virtual room. Given the influence of sequencing effects and the influence of collision anxiety on spatial presence, the results show that collision anxiety is a complex facet of the user experience in VR.

6 Limitations and Future Work

As discussed above, we decided against separating the pace and extent of movement in favor of developing a VR game that is fun to play and offers an easier transfer to real-world results instead of testing different lab scenarios. However, this means that we cannot separate both factors in our results. Future work might find a better solution and provide a clearer differentiation between the pace and extent of movement. We also cannot be sure about actual differences in the players' movements, as we did not track their movements. However, we can be reasonably confident that players did move more in the sword version, as there was a significant difference in the BORG scale between both conditions. In the same vein, while we tried to keep both versions as comparable as possible, the different weapons might still have induce unintended side effects. The differences in the CAQ score depending on the sequence of the game versions hint towards the influence of prior experiences with VR. We used an within-design and a randomized order of the games per participant to minimize confounding variables, but since our sample had relatively little VR expertise, future research could benefit from examining a more experienced sample to see if an intense first game version would also influence more experienced participants. Additionally, it would be valuable to explore how long the collision anxiety experienced during one VR session impacts subsequent VR sessions.

7 Conclusion

Fear of real-world collisions can negatively influence the player experience in VR applications. To understand the relationship between game design and collision anxiety, we developed two versions of a VR game—one slower version with low movement requirements

and one fast-paced version that requires players to make large movements, walk, and turn around. After a pre-study to check for unintended differences between both game versions, 35 participants played both games in a within-design in a randomized order of the games. Results showed that the version, which required less player movement, caused significantly less overall collision anxiety than the faster-paced game version. However, we also observed sequencing effects: The group that played the faster game first reported significantly higher collision anxiety in the slower pistol game variant than the pistol-first group. Hence, while movement requirements seem to play a role in the occurrence of collision anxiety, it seems as if collision anxiety does not only stem from the game experience itself but might also be dependent on user expertise and priming effects from earlier experiences. We hope our results encourage further research in the relatively new field of research to better understand the influence of collision anxiety on the user experience in VR.

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A Collision Anxiety

A.1 CAQ subscales per game version

Table 1: Descriptive Values of the CAQ

Subscale	Group	<i>M</i>	<i>SD</i>
Overall CAQ	Pistol	2.09	0.69
Score*	Sword	3.22	0.61
General Collision	Pistol	1.28	0.40
Anxiety	Sword	2.89	0.96
Orientation	Pistol	2.90	1.25
	Sword	3.55	0.81

* The overall CAQ score consists only of the two subscales general collision anxiety and orientation. Following the advice of the original authors of the CAQ we left out the subscale interpersonal collision anxiety as there was no possibility of interpersonal collisions.

A.2 Descriptive differences between CAQ subscales

Table 2: Descriptive differences between the pistol-first and sword-first group in the CAQ mean score and the two subscales.

CAQ (subscales)	Group	Mean	SD
CAQ Score:	Pistol	1.78	0.49
Pistol	Sword	2.42	0.74
CAQ Score:	Pistol	3.23	0.66
Sword	Sword	3.22	0.58
ORI: Pistol	Pistol	2.26	0.91
	Sword	3.57	1.22
ORI: Sword	Pistol	3.22	0.71
	Sword	3.90	0.77
GCA: Sword	Pistol	3.22	0.98
	Sword	2.53	0.83
GCA: Pistol	Pistol	1.29	0.29
	Sword	1.27	0.50

B iPQ Subscales per Game Version

Table 3: Descriptive Values of the iGroup Presence Questionnaire [18]

Item	Group	<i>M</i>	<i>SD</i>
Presence	Pistol	4.89	1.41
	Sword	5.03	1.25
Spatial Presence	Pistol	5.13	0.95
	Sword	5.39	0.81
Involvement	Pistol	4.06	0.65
	Sword	3.86	0.78
Experienced	Pistol	3.14	1.07
Realism	Sword	3.11	1.05

C PXI Subscales per Game Version

Table 4: Descriptive values of the short version of the Player Experience Inventory [7]

Item	Group	<i>M</i>	<i>SD</i>
Ease of	Pistol	6.51	0.95
Control	Sword	6.94	0.24
Challenge	Pistol	5.29	1.56
	Sword	4.83	2.09
Audiovisual	Pistol	6.20	0.90
Appeal	Sword	6.23	0.88
Curiosity	Pistol	4.91	1.65
	Sword	4.91	1.62
Clarity of	Pistol	6.89	0.32
Goals	Sword	6.91	0.37
Autonomy	Pistol	5.20	1.41
	Sword	5.86	1.17
Mastery	Pistol	5.86	1.51
	Sword	4.97	1.54
Meaning	Pistol	4.06	1.33
	Sword	6.43	0.66
Immersion	Pistol	6.09	0.89
	Sword	5.94	1.64
Progress	Pistol	5.94	1.64
Feedback	Sword	5.80	1.76